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INSTITUTE OF ENGINEERING
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**A MINOR PROJECT REPORT ON AUTOMATED IOT BASED
ATTENDANCE TRACKING**

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DECLARATION

We hereby declare that the project report entitled "**Automated IoT-Based Attendance Tracking**" submitted to the Department of Electronics and Computer Engineering, IOE, Purwanchal Campus, in partial fulfillment of the requirements for the Degree of Bachelor of Engineering in Electronics, Communication, and Information Technology, is the result of our dedicated efforts and hard work.

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ABSTRACT

The **Automated IoT-Based Attendance Tracking** system is designed to automate the process of marking attendance using Radio Frequency Identification (RFID) technology. Traditional attendance methods, such as manual registers, often face challenges like time consumption, human errors, and security vulnerabilities. This project addresses these issues by integrating RFID tags, an RFID reader, and an ESP32-CAM to capture real-time attendance data efficiently.

In this system, each user is assigned a unique RFID tag, which is scanned by an RFID reader to record attendance. The scanned data is processed and transmitted to a server, where it is stored in a database for real-time tracking and verification. To enhance security, an ESP32-CAM captures images during each scan, and face recognition techniques are applied to verify the identity of the user. The system ensures accuracy, security, and reliability, reducing the chances of proxy attendance.

This project demonstrates the practical application of RFID technology in attendance management, offering a cost-effective and efficient solution for schools, colleges, and workplaces. The implementation of face verification alongside RFID scanning further enhances the reliability of the system, making it more robust and tamper-proof.

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LIST OF ABBREVIATIONS

RFID	: Radio-Frequency Identification
API	: Application Programming Interface
Colab	: Colaboratory
Wi-Fi	: Wireless Fidelity
IoT	: Internet of things
UID	: Unique Identifier
IDE	: Integrated Development Environment

CHAPTER 1

INTRODUCTION

1.1 Background

Attendance tracking is a fundamental activity in educational institutions and workplaces. Traditional systems often rely on manual processes that are prone to errors, inefficiencies, and fraud. With advancements in technology, automated solutions IoT are increasingly becoming viable, ensuring both accuracy and ease of use.

1.2 Gap Identification

Currently, attendance tracking systems face challenges in balancing efficiency, accuracy, and ease of use. Traditional methods, such as manual registers is time-consuming, prone to errors.

1.3 Motivation

The motivation for this project stems from the need to modernize and streamline attendance management processes, which are often plagued by inefficiencies in traditional methods. Observing the challenges faced by institutions, such as manual errors, time consumption, and the difficulty of ensuring accuracy, inspired us to create a solution that leverages RFID technology. Our goal is to provide a:

- Reliable, Cost-effective, and User-friendly system
- Incorporating real-time data tracking
- Efficient reporting by offering seamless integration with institutional databases.

By bridging the gaps in traditional systems and leveraging modern technology, our project aims to set a new standard for efficiency, accuracy, and reliability in attendance management.

1.4 Objectives

1.4.1 Primary Objective

- To eliminate manual processes by using RFID technology for quick and accurate attendance recording in real time.

1.4.2 Secondary Objectives

- To manage automatic timetracking and update attendance periodwise.
- Create a webapp that features; daily report generation, administrator and user roles, notifying the absentees for application and so on.

CHAPTER 2

RELATED THEORY

2.1 Project Components

2.1.1 Hardware:

- **RFID Scanner:** An RFID scanner is a device that uses radio waves to read information stored on RFID tags, typically a unique identifier. It consists of a reader and an antenna to detect and communicate with tags, transmitting the scanned data to a connected system for further processing. This technology is commonly used in applications like attendance systems, inventory tracking, and access control.

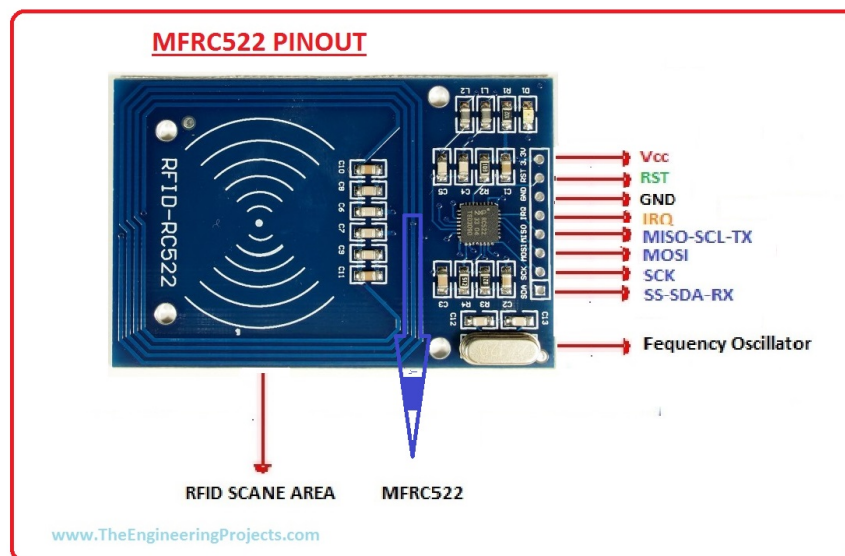


Figure 2.1: RFID SCANNER (MFRC522)

The MFRC522 is a widely used RFID (Radio Frequency Identification) reader module that operates at 13.56 MHz. It supports MIFARE RFID cards and tags, making it ideal for applications like access control, attendance systems, and inventory tracking.

- **RFID Tags :** RFID tags compatible with the MFRC522 module typically operate at 13.56 MHz. These tags contain unique identifiers (UIDs) that can be read by the MFRC522 reader. UIDs are mapped to the data of the students or attendees using SQLite.
- **ESP32 cam:** The heart of the ESP32-CAM is an ESP32-S System-on-Chip (SoC) from Ai-

Thinker. Being an SoC, the ESP32-S chip contains an entire computer—the microprocessor, RAM, storage, and peripherals—on a single chip.

In our System, the ESP32 can serve as the network interface between the RFID scanning system and the Python backend server. It will handle the Wi-Fi communication, allowing the system to send the scanned RFID UID data to the backend server for real-time attendance updates. The ESP32 can be programmed using the Arduino IDE.

The OV2640 camera sensor on the ESP32-CAM is what sets it apart from other ESP32 development boards and makes it ideal for use in our project which can be used for fraud detection.

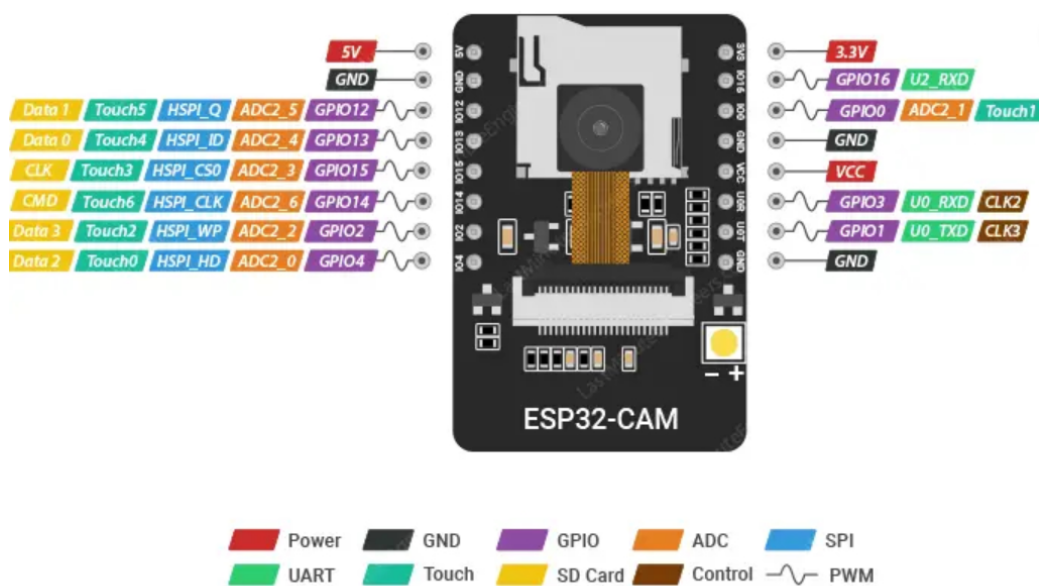


Figure 2.2: ESP32 cam

2.1.2 Software Components

- **SQLite and Flask:** SQLite is a fast, open-source relational database system used to manage and organize data with SQL, popular for web and software development.

Flask is a web framework for Python used to build the server-side application that will handle requests from the Arduino (via ESP32) and the user interface (frontend). Flask makes it easy to manage routing, handle HTTP requests, and interact with a database.

SQLite will be used to store attendance records. Each record includes the UID (from the scanned RFID tag), subject, time, date, period, and so on.

Flask will act as the backend server, receiving data from the Arduino (scanned RFID UID), processing it, and updating the SQLite database. It will also handle user authentication and interactions with the frontend.

- **Arduino IDE:** The Arduino IDE is a software development environment used to write and upload code to Arduino boards (e.g., Arduino Uno, ESP32). In our project, the Arduino IDE will be used to program the ESP32 (which will interface with the RFID reader) to capture RFID tags and send the data to the Flask server for processing.
- **HTML, CSS and JavaScript:** For user authentication, the frontend is built using HTML, CSS, and JavaScript. The user will log in using a username and password, and only after successful authentication will they be able to interact with the attendance dashboard. Flask serves these pages, and Flask-Login handles session management and user authentication.

CHAPTER 3

LITERATURE REVIEW

[1] The transition to RFID-based attendance systems has revolutionized attendance tracking by replacing manual methods with fast, contactless solutions. RFID systems use tags assigned to individuals, which are scanned by readers to log attendance. These systems offer significant advantages, including accuracy, efficiency, and scalability, making them ideal for large-scale applications.

Radio Frequency Identification (RFID) is a popular choice due to its ability to rapidly and contactlessly identify individuals. Studies by Kumar et al. (2021) show that RFID systems reduce the time taken to record attendance while ensuring high accuracy.

[2] IoT systems rely heavily on cloud computing for storing, processing, and analyzing data. Cloud platforms allow remote access to attendance records and facilitate real-time reporting.

[3] Automated IoT-based attendance systems address the limitations of traditional methods by leveraging interconnected devices and cloud integration. These systems use sensors like RFID readers to capture attendance data, transfer it via Wi-Fi or Bluetooth, and store it on cloud platforms for real-time access. IoT provides scalability, accuracy, and efficiency, making it suitable for organizations of all sizes. Studies have shown that IoT-based systems reduce errors and streamline attendance processes compared to manual methods.

CHAPTER 4

METHODOLOGY

The proposed **Automated IoT-Based Attendance Tracking** integrates RFID technology and face recognition to automate attendance tracking. The system captures real-time attendance by scanning RFID tags and verifying user identity using an ESP32-CAM. Attendance data is then processed, stored in a database, and displayed on a web-based dashboard. This system ensures security, accuracy, and efficiency while preventing proxy attendance.

4.1 Data Collection

The first step in developing the system is gathering and managing the necessary data:

- **RFID-based Attendance Data:** Each student or employee is assigned a unique RFID tag. When scanned using an **MFRC522 RFID reader**, the tag's UID is captured and transmitted.
- **Facial Data for Verification:** An **ESP32-CAM module** captures an image during each RFID scan. These images are used to generate face embeddings for identity verification.
- **User Information Database:** A centralized **SQLite** stores RFID UIDs, names, timestamps, and corresponding face embeddings for authentication.

4.2 Data Processing and Cleaning

To ensure accurate attendance tracking, the collected data must be processed and cleaned:

- **RFID Tag Validation:** The system verifies whether the scanned RFID UID exists in the database. If invalid, an error message is generated.
- **Face Image Preprocessing:** The **ESP32-CAM** captures an image, which undergoes pre-processing, including grayscale conversion, resizing, and face detection using **OpenCV** and a pre-trained deep-learning model .
- **Data Storage:** Processed RFID and facial data are structured and stored in the database for real-time tracking and retrieval.

4.3 System Architecture and Hardware Integration

The system integrates both hardware and software components:

4.3.1 Hardware Components

- **ESP32:** Controls RFID scanning and image capturing.
- **MFRC522 RFID Module:** Scans RFID tags and sends UID data.
- **ESP32-CAM:** Captures images for face verification.
- **Power Supply and Communication:** The ESP32 is connected via Wi-Fi to transmit data to a remote server.

4.3.2 Software Components

- **Embedded Programming:** Written in C++ using the Arduino IDE, handling RFID scanning, ESP32 communication, and HTTP requests.
- **Server & Database:** A Flask-based backend with SQLite/MySQL stores attendance data and performs facial recognition.
- **Web Dashboard:** A front-end interface built with HTML, CSS, and JavaScript enables users to manage attendance records.

4.4 Block Diagram

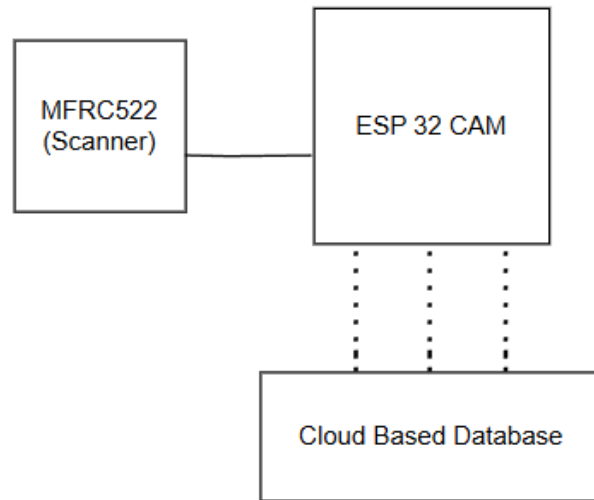


Figure 4.1: Block diagram

4.5 Flowchart

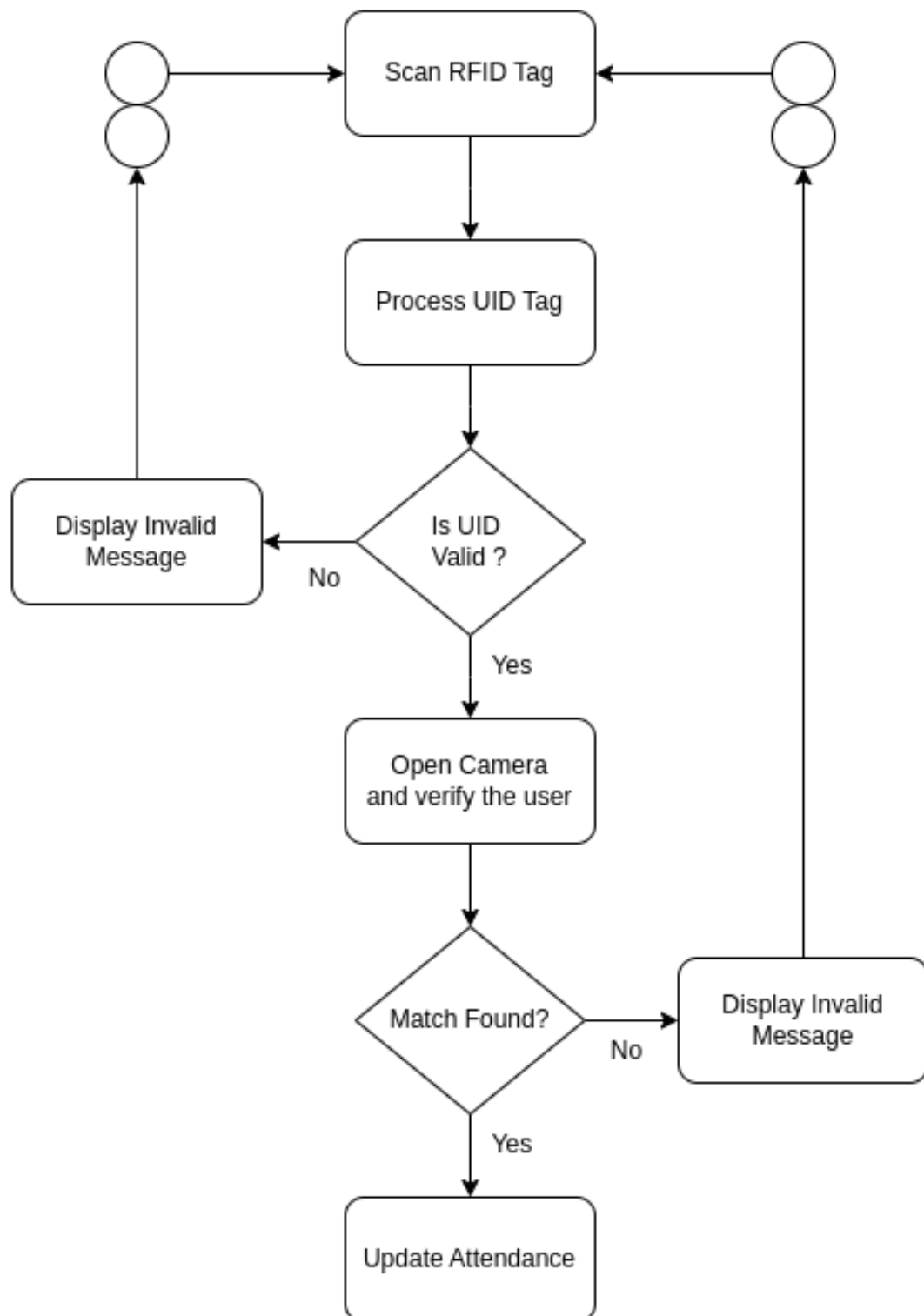


Figure 4.2: Flowchart

CHAPTER 5

EXPECTED RESULTS

The **Automated IoT-Based Attendance Tracking** will provide an automated and secure attendance management solution by combining RFID technology, microcontrollers, and web technologies. The key outcomes include:

- **Automated Attendance Tracking:** RFID tags are scanned using the MFRC522 module, and attendance is recorded in real-time.
- **Seamless Hardware-Software Integration:** ESP32cam handle data collection and transmission, while the Flask server processes and stores attendance data in an SQLite database.
- **Real-Time Data Updates:** Attendance data is updated instantly and is accessible to authenticated users via a web interface.
- **Secure Authentication:** Only authorized users (teachers/admins) can log in to manage or view attendance records, ensuring data security.
- **User-Friendly Frontend:** A web-based dashboard built with HTML, CSS, and JavaScript provides an intuitive interface for viewing and managing attendance data.
- **Scalability and Flexibility:** The modular design allows for future enhancements like analytics, notifications, or expanded database support.

This system will streamline attendance processes, reduce manual effort, and ensure accurate record-keeping in real time.

CHAPTER 6

CONCLUSION

The **Automated IoT-Based Attendance Tracking** is designed to simplify and automate the process of attendance management. By using RFID technology, the system allows students or attendees to scan their RFID tags, which are instantly processed in real time. The scanned data is securely transmitted through microcontrollers like Arduino Uno and ESP32 to a backend server powered by Flask, where it is stored in an SQLite database.

A web-based dashboard, developed with HTML, CSS, and JavaScript, provides an intuitive and user-friendly interface for authorized users, such as teachers or administrators, to view and manage attendance records. The system ensures data security through authentication, allowing only verified users to access or update records. Its modular design also makes it scalable, enabling easy integration of additional features like analytics, notifications, or advanced databases.

By effectively integrating hardware and software components, this system reduces manual effort, eliminates errors, and ensures reliable and accurate attendance tracking, offering a practical and efficient solution for educational institutions or similar environments.

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- [3]Jamil Y. Khan and Mehmet R. Yuce”**Internet of Things (IoT): Systems and Applications**”

APPENDIX A

APPENDIX B